LIBBY — PRECLINICAL HORIZON

aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun2

Preclinical recommendations — forward-looking horizon scan

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Preclinical recommendations

A forward-looking horizon scan of candidate drugs, compounds, and treatment strategies that are earlier than clinical development, or not yet developed, but plausibly target this patient's stated features. These are research directions, not enrollable options or treatment recommendations.

1. CD123-targeted degrader-antibody conjugates with a GSPT1-degrading payload (ORM-1153; Hdz-C123A) (ADC)

Target: CD123-positive blasts (payload chosen to work despite TP53 aberration) **Stage:** IND-enabling **Evidence strength:** Emerging

Rests on sponsor abstracts alone, but it is the one candidate here whose payload does not need the p53 arm this clone has probably lost.

The CD123 agent already on her trial list, pivekimab sunirine, kills through a DNA-alkylating payload whose death signal needs p53 and is gated by the BCL-2 family. That is exactly the wiring a TP53-aberrant, venetoclax-refractory clone has broken. Swapping the payload for a GSPT1 degrader keeps the CD123 handle and drops the p53 requirement, which is why this sits at rank 1. Two groups arrived at the same design independently, ORM-1153 and Hdz-C123A, and that convergence is worth more than either molecule's own data. The data are the weak part: every result is a sponsor conference abstract, none of it peer-reviewed, and the TP53-independence claim comes from isogenic cell lines rather than TP53-aberrant patient material. Her CD123 is qualitative flow only, so antigen density, which is what governs how much payload actually reaches a blast, has never been measured in her.

Mechanism. An anti-CD123 antibody carries a cereblon-based molecular glue that degrades the translation termination factor GSPT1 once the conjugate is internalized. Killing runs through collapse of protein synthesis rather than through DNA damage, which is where p53 would normally be needed.

Translatability. High for this list: potency held in TP53-mutant isogenic AML lines, primary blasts and xenografts, though no TP53-aberrant patient material has been tested.

Developability. Furthest along here. Regulatory submission has been signaled for late 2026, so first-in-human is plausibly within a year. Still nothing accessible today.

Counter-productive MoA. Low (CD123 is shared with normal progenitors, so myeloid recovery ahead of a second transplant could be delayed.)

Key risks.

- Every result is a sponsor conference abstract; nothing is peer-reviewed and no independent group has replicated ORM-1153 itself
- Systemic GSPT1 degraders such as CC-90009 caused hypotension and infusion reactions in AML trials, and antibody conjugation is an untested fix in humans
- CD123 antigen density drives payload delivery and has never been quantified on her blasts
- Whether any CD123 agent reaches a thigh myeloid sarcoma at cytotoxic concentrations is unknown
- No study exists yet, so nothing here can be accessed now

Open questions.

- What is her quantitative CD123 antigen density, and is it above whatever threshold a degrader payload needs?
- Does p53-independent killing hold in TP53-aberrant patient blasts, or only in isogenic lines?
- Does conjugation actually widen the GSPT1 therapeutic index in humans, or only in mice?

Key references. doi:10.1182/blood-2025-5051; doi:10.1158/1538-7445.am2026-1710; doi:10.1182/blood-2024-201221

2. Decitabine plus an ATR inhibitor, as a TP53-mutant-selective combination (combination)

Target: TP53 aberration, pending confirmation, after venetoclax failure **Stage:** In vivo (animal) **Evidence strength:** Moderate

One laboratory, April 2026, unreproduced, and the whole effect hangs on decitabine specifically rather than on hypomethylation. Both drugs already exist, which is the unusual part.

This is the one TP53 idea here that does not need p53 protein working again, which matters a great deal when the variant is unresolved and may turn out to be a copy loss rather than a missense. Baeten's group built the effect in isogenic TP53-mutant AML cells, then cut leukemia burden and lengthened survival in mice. The load-bearing detail is a negative one: azacitidine did not reproduce it, and neither did the DNMT1 inhibitor GSK3685032 at comparable global hypomethylation, so the effect belongs to decitabine specifically and not to demethylation. That specificity is also the most fragile thing on the row. One group, published April 2026, and nobody has had time to reproduce it. Both halves are obtainable drugs, which is unusual on this list. A nearby idea was checked and set aside for the opposite reason: CK1alpha degraders exploit her del(5q) neatly, but they kill largely by stabilizing p53, the arm this clone has probably lost, so that route would push against the goal here. And if the 2026 negative panel is real rather than a limit-of-detection artifact, the selectivity argument does not apply to her at all.

Mechanism. Decitabine slows replication fork progression and produces single-strand breaks, which switches on the ATR checkpoint. TP53-mutant AML cells resolve that stress poorly, so adding an ATR inhibitor pushes them into mitotic catastrophe while TP53-wild-type cells survive.

Translatability. Moderate: isogenic TP53-mutant AML lines plus mouse survival, but one laboratory only, and nothing in patient material or in extramedullary disease.

Developability. Closest to reach on paper: decitabine is approved and ATR inhibitors are in trials, so this is a combination-trial question rather than a discovery one.

Counter-productive MoA. Moderate (Overlapping myelosuppression in a heavily pretreated post-transplant marrow carrying roughly 85% blasts.)

Key risks.

- Her TP53 status is unconfirmed in both directions; a genuinely TP53-wild-type clone removes the selectivity the whole combination rests on
- Effect sizes come from one group and the April 2026 report has not been reproduced
- The decitabine-versus-azacitidine distinction is the load-bearing claim and rests on a single set of experiments

- Hypomethylating-agent exposure during the 2019-2020 MDS phase is undocumented, and decitabine refractoriness would undercut the pairing
- Mouse marrow models say nothing about a thigh myeloid sarcoma

Open questions.

- Does an independent group reproduce the decitabine-specific synergy, or does azacitidine work after all?
- Was she exposed to a hypomethylating agent during the MDS phase, and did she respond?
- What ATR inhibitor dose is tolerable alongside decitabine in a post-transplant marrow?

Key references. pmid:41610325; pmid:38228616

3. Multiplex epitope-edited donor HSPC graft (CD123 / FLT3 / KIT), shielding normal hematopoiesis from antigen-directed therapy (cell therapy)

Target: CD123-positive and CD33-positive blasts, with the donor graft engineered to tolerate antigen-directed therapy **Stage:** In vivo (animal) **Evidence strength:** Moderate

Only worth anything if a second transplant happens and a CD123 or CD33 agent exists to dose above current limits. Two laboratories, no human yet.

Both of her confirmed handles, CD123 and CD33, are shared with normal myeloid progenitors, and that overlap is what caps every CD123 and CD33 agent on her trial list. If a second transplant goes ahead, the graft is where the problem can be solved rather than tolerated. Casirati's group edited CD123, FLT3 and KIT in human HSPCs at 75% to 90% efficiency without altering protein function, and dual-edited cells survived a bispecific CAR-T that cleared AML; a second laboratory reached the same conclusion on the CD123 arm alone. Two things hold it at rank 3. Editing does nothing to the leukemia, so it only pays off paired with an antigen-directed agent dosed above current tolerability, and no such agent is approved. And the whole idea is contingent on HCT2 going forward, which is unsettled in this case. Manufacturing runs in weeks; her peripheral blasts went from 58% to 82% in about a month.

Mechanism. Base editing changes a handful of residues in the antibody epitope of CD123, FLT3 or KIT on donor stem cells without disturbing protein function. Edited hematopoiesis then survives a CAR-T or antibody aimed at the unedited antigen, so the antigen-directed agent can be dosed to leukemia-clearing levels.

Translatability. Moderate: humanized mouse reconstitution and AML xenografts under CAR-T pressure, replicated by a second laboratory on CD123, but no human has received a multiplex-edited graft.

Developability. Platform, not a drug. The CD33 single-edit version reached the clinic, so the path is walkable; this still needs an IND and a paired agent.

Counter-productive MoA. Moderate (Weeks of manufacturing while blasts climb, and the shield gives nothing unless a paired antigen-directed agent is dosed alongside.)

Key risks.

- Requires a second transplant to be feasible, which is its own unresolved question here
- Base editing carries bystander-edit and off-target liabilities that are managed rather than eliminated
- Manufacturing timelines do not match how fast her blast count is moving

- The shield is useless without a CD123 or CD33 agent dosed above current tolerability, and none is approved
- Editing three antigens at once has never been taken into a human
- Needs a sibling donor able and willing to donate a second time

Open questions.

- Does HCT2 go forward, and is the sibling donor available again?
- Which antigen-directed agent would be paired with the edited graft, and at what dose?
- Does multiplex editing keep the single-edit safety profile that reached the clinic?

Key references. pmid:37648862; pmid:37773046; pmid:31138698

4. 753B, a VHL-recruiting dual BCL-xL/BCL-2 PROTAC degrader (PROTAC / degrader)

Target: TP53 aberration with venetoclax failure and compensatory BCL2L1 reliance **Stage:** In vivo (animal) **Evidence strength:** Moderate

Lines up with the resistance mechanism her phenotype suggests, but the primary-sample data never state TP53 status, so the match is inferred rather than shown.

Her disease progressed on venetoclax with a CD34-positive, CD14-negative, CD64-negative phenotype, which argues against the monocytic MCL1 escape route and toward the BCL-xL arm that Nechiporuk mapped to TP53, BAX or PMAIP1 loss. 753B destroys BCL-xL and BCL-2 rather than occupying their grooves, which matters more if the clone has raised BCL2L1 protein levels. Platelets lack VHL, and that trick did hold in mice. One group produced the entire package, and the TP53 status of the responsive primary samples was never reported, so the fit to a TP53-aberrant clone is assumed rather than shown. That is what keeps it out of the top three. The senolytic claim in the same paper is a separate hypothesis and adds nothing to the venetoclax-resistance argument.

Mechanism. 753B ties BCL-xL and BCL-2 to the von Hippel-Lindau E3 ligase so both proteins are ubiquitinated and destroyed rather than occupied. Platelets do not express VHL, which is the design trick meant to separate 753B from navitoclax on thrombocytopenia.

Translatability. Moderate: AML lines, venetoclax-resistant primary samples and PDX models, but the TP53 status of responders is unreported, so the fit here is inferred.

Developability. Academic compound, no IND. The VHL-sparing concept reached the clinic as DT2216, so the regulatory path is mapped even though 753B is not dosable.

Counter-productive MoA. Moderate (Degrading BCL-2 as well brings back the same on-target myelosuppression venetoclax already caused in this patient.)

Key risks.

- Single-group preclinical package with no independent replication
- The TP53 status of the responsive primary samples is unreported, so the match to her clone is assumed
- Platelet sparing via VHL held in mice; whether it survives active human exposures is untested
- Blasts must express enough VHL for the degrader to work, and VHL levels in AML are not routinely measured

Open questions.

- Do TP53-aberrant primary samples respond to 753B specifically?
- Does the platelet-sparing margin hold at exposures that degrade BCL-xL in marrow?
- Would DT2216 clinical data settle the safety question before 753B ever needs its own IND?

Key references. pmid:37078252; pmid:31048320

5. Whole-exome minor-histocompatibility-antigen mapping of this recipient/sibling-donor pair to nominate non-HA-1 GvL targets (conceptual strategy)

Target: HLA-restricted immunotherapy handles, HLA-A*02:01 unretrieved **Stage:** Early translational **Evidence strength:** Emerging

Decision support rather than a therapy: it widens the target search if A*02:01 comes back negative, but a nominated antigen is still not a product.

Every cellular option this case was submitted for sits behind HLA-A*02:01, and the allele-level result from the 2020 transplant record has never been retrieved. If it comes back negative the whole stack falls at once, leaving a single A*24:02-restricted WT1 product as the fallback. Cieri and colleagues built a pipeline that finds minor histocompatibility antigens straight from donor and recipient exomes, validated it across 220 matched pairs, and confirmed nominated graft-versus-leukemia targets in a further 58. Run on this specific sibling pair, it converts a binary eligibility question into a search. The ceiling is that the output is a list of candidate antigens, not a therapy. Building a TCR against a bespoke target takes years, the discovery timeline does not match blasts that went from 58% to 82% in a month, and it needs retrievable germline DNA from the 2020 donor that may no longer exist.

Mechanism. Whole-exome sequencing of germline DNA from donor and recipient finds the coding polymorphisms where they differ, then class I binding prediction, ligandome detection and expression filtering narrow those to peptides a donor T cell could see on hematopoietic tissue. The output is a ranked list of candidate GvL targets restricted to whatever HLA alleles this pair actually carries.

Translatability. Moderate: validated across 220 donor-recipient pairs with a 58-pair confirmation cohort, but the work nominates targets rather than showing one was successfully drugged.

Developability. The sequencing half is commissionable today from stored donor and recipient DNA. Turning a nominated antigen into an actual product takes years.

Counter-productive MoA. Moderate (A nominated antigen with broad tissue expression drives graft-versus-host disease instead of the graft-versus-leukemia effect it was picked for.)

Key risks.

- Requires retrievable germline DNA from the 2020 sibling donor, which may not exist
- Discovery timelines do not match how fast this disease is moving
- A nominated antigen is not a product; no clinical route exists for a bespoke target
- Escape by antigen loss or class I downregulation applies here exactly as it does to HA-1

Open questions.

- Is banked germline DNA from the 2020 sibling donor still available?
- Would a nominated non-HA-1 antigen have any route to a clinical product on a timescale that matters here?
- Does the expression filter separate GvL from GvHD targets reliably enough to act on?

Key references. pmid:39169264

6. Direct BAX activator BTSA1 (BAX trigger-site chemical probe) (small molecule)

Target: TP53 aberration, the apoptotic effector arm sitting below p53 **Stage:** In vivo (animal) **Evidence strength:** Moderate

Mechanistically the cleanest route past a broken p53 arm, and also the least developed: one laboratory, 2017, unreplicated, with no clinical-grade molecule anywhere.

Her blasts grew through venetoclax, and Nechiporuk's screen makes TP53 loss on its own enough to produce that phenotype, because the p53 target genes supplying BH3-only activators go missing. BTSA1 starts one step below where the wiring is cut, binding BAX directly rather than waiting for transcription. Reyna reported activity in the CD34+CD38- fraction of high-risk AML, the compartment that survived FLAG-IDA here. Everything under that is thin. The in vivo result comes from one laboratory in 2017, nobody has independently reproduced it, and eight years on there is still no IND-enabling BAX activator, which for a target this attractive says something about the chemistry. No published data exist in TP53-mutant AML specifically, so the p53-bypass claim is inferred rather than tested. And if this clone took the BAX-null route to venetoclax resistance, the drug has no substrate at all.

Mechanism. BTSA1 binds the N-terminal trigger site of BAX and forces the conformational change that drives BAX translocation and oligomerization at the outer mitochondrial membrane. Because it engages BAX protein directly, it does not wait on p53 to transcribe PUMA or NOXA.

Translatability. Low: AML lines, primary blasts including CD34+CD38- cells, and a xenograft, all from one laboratory, none of it in TP53-mutant AML specifically.

Developability. Chemical probe only. Eight years after the paper there is no IND-enabling BAX activator, so reaching a patient means starting medicinal chemistry from scratch.

Counter-productive MoA. Moderate (Needs intact BAX; a BAX-null resistance route leaves no substrate, and free BCL-2 or BCL-xL re-sequesters what it activates.)

Key risks.

- Single-laboratory 2017 in vivo result with no independent replication in eight years
- Needs intact BAX protein; a BAX-null resistance route leaves nothing for the drug to act on
- Potency is capped by how much free BCL-2 and BCL-xL is available to re-sequester activated BAX
- BAX is ubiquitous, and on-target apoptosis in normal hematopoiesis and gut has never been characterized in humans
- No published data in TP53-mutant AML, so the p53-bypass argument is inferred

Open questions.

- Is BAX protein intact in her blasts?
- Has any group reproduced the 2017 xenograft result, in TP53-mutant AML or otherwise?
- Would BAX activation be tolerable in normal tissue at doses that kill blasts?

Key references. pmid:29017059; pmid:31048320

7. CD33 C2-set-domain-directed binders and CAR T cells (CD33 PAN antibodies) (cell therapy)

Target: CD33-positive blasts, antigen density and splice genotype unmeasured **Stage:** In vivo (animal)
Evidence strength: Emerging

Answers the splice-genotype question rather than the antigen-density one, and no clinical-grade C2-directed product exists anywhere today.

Her CD33 is qualitative flow only, her rs12459419 genotype is unknown, and the profile already asks for quantitative antigen density before anyone commits to CD33-directed therapy. A C2-directed binder makes the splice genotype irrelevant and lowers the density needed to fire the receptor, so it addresses two open questions with one construct. Worth recording what did not survive scrutiny alongside it: the neighboring idea of targeting the exon-2-deleted isoform itself was chased and closed. Isoform-specific antibodies found CD33delE2 transcript widespread but no surface protein on any AML line or on primary blasts from 21 patients, so the splice-escape target does not appear to reach the membrane. The C2 binder is kept for a different and better reason, synapse geometry and density rather than escape. It is still a Fred Hutch research program with no clinical-grade vector, and changing epitope does not change how much CD33 sits on a blast, so a genuinely CD33-low clone resists either way.

Mechanism. Every CD33 agent in clinical use binds the membrane-distal V-set domain, which disappears when the rs12459419 splicing SNP drives exon-2 skipping. Binders raised against the membrane-proximal C2-set domain recognize CD33 whether or not the V-set domain is present, and the shorter binding distance also excludes CD45 from the immune synapse, which sharpens CAR T-cell signaling.

Translatability. Low: engineered CD33-isoform lines, primary-sample antibody binding and cytotoxicity assays. A C2-directed bispecific reached phase 1 elsewhere and was discontinued.

Developability. Academic constructs with no clinical-grade vector or IND. Its nearer-term value is as an argument for ordering the quantitative CD33 flow and splice genotype.

Counter-productive MoA. Moderate (On-target myelotoxicity matches every other myeloid CAR and could compromise the marrow a second transplant depends on.)

Key risks.

- Changing epitope does not change total CD33 on the blast, so a genuinely CD33-low clone still resists
- CD33 CAR T cells carry the same prolonged myelotoxicity as every other on-target-off-tumor myeloid CAR
- No clinical-grade C2-directed CAR product exists anywhere
- The companion splice-escape argument was tested and did not hold, so the case rests on density and geometry alone

Open questions.

- What is her quantitative CD33 antigen density, and what is her rs12459419 genotype?
- Does the synapse-geometry advantage translate into a lower density threshold in primary blasts?
- Why was the C2-directed bispecific discontinued, and does that reason apply to a CAR?

Key references. pmid:33589747; pmid:39224504; pmid:32071429

8. EZH2 PROTAC degrader (MS177 class) to reverse non-genomic HLA class II silencing at post-transplant relapse (PROTAC / degrader)

Target: Post-transplant immune escape upstream of any HLA-restricted cellular therapy **Stage:** In vitro
Evidence strength: Speculative

The del(7q) tension is unresolved and points the wrong way: she may be exactly the wrong patient for degrading whatever EZH2 she has left.

Her relapse six years after a matched-sibling transplant looks like graft-versus-leukemia escape rather than graft failure, and her flow shows HLA-DR variably decreased. If class II loss at post-transplant relapse is PRC2-enforced, reopening it before donor cells go in is a sequencing question worth asking, and a degrader removes the non-canonical EZH2-cMyc arm that catalytic inhibitors leave running. The reason this sits at rank 8 rather than higher is specific and should not be smoothed over: EZH2 sits at 7q36.1 and she carries del(7q), so the gene may already be hemizygous in her clone, and EZH2 loss behaves as a myeloid tumor suppressor. There is a real chance she is the wrong patient for this mechanism entirely. Nobody has measured whether her blasts have lost class II either; HLA-DR variably decreased on a diagnostic flow panel is not that measurement. The antigen-presentation evidence is myeloma transcriptomics plus a preprint in a different AML subtype that used a catalytic inhibitor rather than a degrader.

Mechanism. MS177 pulls EZH2 to cereblon and degrades it, taking down both the canonical EZH2-PRC2 complex and the non-canonical EZH2-cMyc complex that catalytic inhibitors leave running. Removing the protein rather than blocking the SET domain reopens PRC2-closed loci faster and, in the myeloma work, switched on interferon-gamma response and MHC antigen-processing genes.

Translatability. Low, cross-tumor only: the antigen-presentation readout is myeloma, and the leukemia evidence is a preprint in FUS::ERG AML using a catalytic inhibitor.

Developability. No EZH2 degrader has entered IND-enabling work. The accessible substitute is an approved catalytic inhibitor, which this same literature argues is the weaker tool.

Counter-productive MoA. High (EZH2 sits at 7q36.1 and she carries del(7q); further loss may drive the myeloid clone rather than restrain it.)

Key risks.

- EZH2 sits at 7q36.1 and she carries del(7q), so the gene may already be hemizygous and further loss could favor the clone rather than restrain it
- Nobody has measured class II loss on her blasts; HLA-DR variably decreased on a diagnostic panel is not that measurement
- The class II restoration data are myeloma transcriptomics plus a preprint in a different AML subtype
- Reopening antigen presentation does nothing without a donor effector population present, so it only makes

sense bolted to DLI or a second transplant
- Restoring class II would also raise the graft-versus-host risk it is trying to exploit

Open questions.

- Is EZH2 hemizygous in her clone, and does the del(7q) breakpoint include 7q36.1?
- Have her blasts actually lost HLA class II, measured properly rather than inferred from a diagnostic panel?
- Would class II restoration without a donor effector population do anything at all?

Key references. pmid:35210568; pmid:36747009; biorxiv:10.1101/2024.05.14.594150

9. Non-rezatapopt Y220C chemotypes: covalent azaindoles (KG13) and the PK7088/PK9318 carbazole stabilizer series (small molecule)

Target: TP53 Y220C, contingent on variant confirmation **Stage:** In vitro **Evidence strength:** Speculative

A fallback only if the variant is confirmed Y220C and rezatapopt is off the table. Neither chemotype has been tested in a myeloid cell.

Rezatapopt is the only Y220C reactivator with a myeloid program and it already sits in the clinical track, so this is a fallback chemotype question rather than a parallel option. If her variant does turn out to be Y220C and rezatapopt is unavailable or declined, two structurally distinct routes to the same pocket exist in the academic literature: the PK7088 carbazole series holds the crevice non-covalently, while KG13 reacts covalently with the cysteine the substitution creates. Both are early. KG13 has a 1.7 angstrom structure and thermal-stability data; the carbazole series restored p53 target-gene transcription in a Y220C solid-tumor line. Neither has been run in a leukemia cell. The row also evaporates unless the pending NGS returns Y220C specifically, and her TP53 status is currently unresolved in both directions: a 2019 pathogenic call against a 2026 negative on a panel reading to 2% VAF with no copy-number calling. That is a weak footing to rank anything on.

Mechanism. The Y220C substitution opens a surface crevice next to a free cysteine. The carbazole series occupies that crevice non-covalently and raises the melting temperature of the mutant protein; KG13 instead reacts covalently with the mutant cysteine, which locks in wild-type thermal stability under a different selectivity logic.

Translatability. Low: recombinant protein crystallography and thermal shift, plus one Y220C solid-tumor line. Nothing in myeloid cells, and the variant itself is unconfirmed here.

Developability. Structure-biology probes. Neither series has cellular potency good enough to justify IND work, and rezatapopt supersedes both wherever it is accessible.

Counter-productive MoA. Moderate (Restored p53 needs a BCL-2-family partner to kill in AML, and she has already progressed through venetoclax.)

Key risks.

- The whole idea evaporates unless the pending NGS returns Y220C specifically
- Her TP53 status is unresolved in both directions, so even the parent feature is not established
- No myeloid model data, so the AML-specific point that reactivated p53 needs a BCL-2-family partner to kill is untested for these chemotypes

- Covalent cysteine chemistry carries a proteome-selectivity burden that KG13 has not resolved

Open questions.

- What was the 2019 variant identity and VAF, and does p53 IHC or 17p FISH settle the discrepancy?
- Would either chemotype restore p53 function in myeloid cells, where nobody has tried?
- Does reactivated p53 kill at all in a clone that has already escaped venetoclax?

Key references. pmid:36197521; pmid:23630318

10. DRP1 inhibition (Mdivi-1) to restore venetoclax-induced mitochondrial apoptosis in TP53-mutant AML (repurposed drug)

Target: TP53 aberration with venetoclax refractoriness **Stage:** In vitro **Evidence strength:** Speculative

A single in-vitro paper with a tool compound of contested selectivity, kept only so the apoptosis axis is complete. Ranked last deliberately.

This is the weakest row kept, and it is here for completeness of the apoptosis-restoration axis rather than because the evidence carries it. The mechanism is on-axis: lowering the BAX and BAK activation threshold from the mitochondrial-dynamics side puts it alongside the BAX-activator and BCL-xL entries above, which is exactly where it belongs and no higher. The support is one 2023 paper in TP53-mutant AML cell lines. No animal data, no independent replication I could find, and Mdivi-1's selectivity for DRP1 has been challenged in the wider literature, with some of its cellular effects attributed to complex I inhibition instead. The premise is also re-challenge with venetoclax after documented progression, which is a poor starting point. Mdivi-1 is a laboratory reagent; no clinical-grade DRP1 inhibitor exists and no sponsor is obviously interested in making one.

Mechanism. Blocking the mitochondrial fission GTPase DRP1 shifts mitochondrial morphology in a way that lowers the threshold for BAX and BAK activation, which the authors report restores venetoclax-driven outer-membrane permeabilization in TP53-mutant cells.

Translatability. Low: TP53-mutant AML cell lines only, one paper, no in vivo work, and the tool compound's DRP1 selectivity has been publicly questioned.

Developability. Furthest from a patient. Mdivi-1 is a laboratory reagent, there is no clinical-grade DRP1 inhibitor, and no obvious sponsor for one.

Counter-productive MoA. High (Mitochondrial fission is essential in normal tissue, the window is uncharacterized, and off-DRP1 effects may drive the phenotype.)

Key risks.

- Rests on a single in-vitro paper with no in-vivo confirmation and no independent replication
- Mdivi-1's specificity has been challenged; some of its cellular effects have been attributed to complex I inhibition rather than DRP1
- The premise is re-challenging with venetoclax after documented progression
- Mitochondrial fission machinery is essential in normal tissue and the therapeutic window is uncharacterized

Open questions.

- Does the effect survive a genetic DRP1 knockdown rather than Mdivi-1?
- Has anyone reproduced it in vivo, or in primary TP53-aberrant blasts?
- Is there any reason to expect venetoclax re-challenge to work after documented progression?

Key references. pmid:36765703